

DESIGN CODE OF WALLECE TREE MULTIPLIER

```
/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/rtl/lab3_wallace.vhd [Verilog]
```

```
1  -----
2  --
3  -- Company:
4  -- Engineer:
5  --
6  -- Create Date: 09/03/2026 04:51:57 PM
7  -- Design Name:
8  -- Module Name: lab3_wallace - Behavioral
9  -- Project Name:
10 -- Target Devices:
11 -- Tool Versions:
12 -- Description:
13 --
14 -- Dependencies:
15 --
16 -- Revision:
17 -- Revision 0.01 - File Created
18 -- Additional Comments:
19 -----
20 --
21
22 library IEEE;
23 use IEEE.STD_LOGIC_VECTOR.ALL;
24
25 -- Uncomment the following library declaration if using
26 -- arithmetic functions with Signed or Unsigned values
27 --use IEEE.NUMERIC_STD.ALL;
28
29 -- Uncomment the following library declaration if instantiating
30 -- any Xilinx leaf cells in this code.
31 --library UNISIM;
32 --use UNISIM.VComponents.all;
33
34 entity lab3_wallace is
35     Port ( a : in STD_LOGIC_VECTOR (3 downto 0);
36           b : in STD_LOGIC_VECTOR (3 downto 0);
37           c : out std_logic_vector (7 downto 0));
38 end lab3_wallace;
39
40 architecture Behavioral of lab3_wallace is
41
42     component rca8 is
43     Port ( a : in STD_LOGIC_VECTOR (7 downto 0);
44           b : in STD_LOGIC_VECTOR (7 downto 0);
45           sum : out STD_LOGIC_VECTOR (7 downto 0);
```

46	cout : out STD_LOGIC);
47	end component;
48	
49	signal p0 : std_logic_vector(7 downto 0);
50	signal p1 : std_logic_vector(7 downto 0);
51	signal p2 : std_logic_vector(7 downto 0);
52	signal p3 : std_logic_vector(7 downto 0);
53	
54	signal r0 : std_logic_vector(7 downto 0);
55	signal r1 : std_logic_vector(7 downto 0);
56	signal r2 : std_logic_vector(7 downto 0);
57	
58	signal r3 : std_logic_vector(7 downto 0);
59	signal r4 : std_logic_vector(7 downto 0);
60	
61	signal cout : std_logic;
62	
63	begin
64	process(a , b , p0 , p1 , p2 , p3 , r0 , r1 , r2 , r3 , r4)
65	
66	begin
67	
68	p0 <= (others => '0');
69	p1 <= (others => '0');
70	p2 <= (others => '0');
71	p3 <= (others => '0');
72	
73	for i in 0 to 3 loop
74	p0(i) <= a(0) and b(i);
75	p1(i+1) <= a(1) and b(i);
76	p2(i+2) <= a(2) and b(i);
77	p3(i+3) <= a(3) and b(i);
78	end loop;
79	
80	r0 <= (others => '0');
81	r1 <= (others => '0');
82	r2 <= (others => '0');
83	r3 <= (others => '0');
84	r4 <= (others => '0');
85	
86	r0(0) <= p0(0);
87	r0(1) <= p0(1) xor p1(1);
88	r0(2) <= p0(2) xor p1(2) xor p2(2);
89	
90	r0(3) <= p0(3) xor p1(3) xor p2(3);
91	
92	r0(4) <= p3(4) xor p2(4) xor p1(4);
93	r0(5) <= p3(5) xor p2(5);
94	r0(6) <= p3(6);
95	r0(7) <= '0';

96	
97	
98	
99	r2(3) <= p3(3);
100	
101	r1(0) <= '0';
102	r1(1) <= '0';
103	r1(2) <= p0(1) and p1(1);
104	r1(3) <= (p0(2) and p1(2)) or (p1(2) and p2(2)) or (p2(2) and p0(2)
105	);
106	r1(4) <= (p0(3) and p1(3)) or (p1(3) and p2(3)) or (p2(3) and p0(3)
107	);
108	r1(5) <= (p1(4) and p2(4)) or (p2(4) and p3(4)) or (p3(4) and p1(4)
109	);
110	r1(6) <= p2(5) and p3(5);
111	r1(7) <= '0';
112	
113	r3(0) <= r0(0);
114	r3(1) <= r0(1);
115	r3(2) <= r0(2) xor r1(2);
116	r3(3) <= r0(3) xor r1(3) xor r2(3);
117	r3(4) <= r0(4) xor r1(4);
118	r3(5) <= r0(5) xor r1(5);
119	r3(6) <= r0(6) xor r1(6);
120	r3(7) <= r0(7) xor r1(7);
121	
122	r4(0) <= '0';
123	r4(1) <= '0';
124	r4(2) <= '0';
125	r4(3) <= r0(2) and r1(2);
126	r4(4) <= (r0(3) and r1(3)) or (r1(3) and r2(3)) or (r2(3) and r0(3)
127	);
128	r4(5) <= r0(4) and r1(4);
129	r4(6) <= r0(5) and r1(5);
130	r4(7) <= r0(6) and r1(6);
131	
132	end process;
133	
134	final_add : rca8 port map (a => r3 , b => r4 , sum => c , cout => cout);
135	end Behavioral;

TEST BENCH CODE OF CLA

/home/dhruv/cao_lab/vivado_proj/cao_lab_u24ev057/tb/lab3_wallace_tb.vhd [Verilog]	
1	-----
	--

2	-- Company:
3	-- Engineer:
4	--
5	-- Create Date: 09/03/2026 05:37:05 PM
6	-- Design Name:
7	-- Module Name: lab3_wallance_tb - Behavioral
8	-- Project Name:
9	-- Target Devices:
10	-- Tool Versions:
11	-- Description:
12	--
13	-- Dependencies:
14	--
15	-- Revision:
16	-- Revision 0.01 - File Created
17	-- Additional Comments:
18	--
19	-----
20	--
21	
22	library IEEE;
23	use IEEE.STD_LOGIC_1164.ALL;
24	
25	-- Uncomment the following library declaration if using
26	-- arithmetic functions with Signed or Unsigned values
27	--use IEEE.NUMERIC_STD.ALL;
28	
29	-- Uncomment the following library declaration if instantiating
30	-- any Xilinx leaf cells in this code.
31	--library UNISIM;
32	--use UNISIM.VComponents.all;
33	
34	entity lab3_wallance_tb is
35	-- Port ();
36	end lab3_wallance_tb;
37	
38	architecture testing of lab3_wallance_tb is
39	
40	component lab3_wallace is
41	Port (a : in STD_LOGIC_VECTOR (3 downto 0);
42	b : in STD_LOGIC_VECTOR (3 downto 0);
43	c : out STD_LOGIC_VECTOR (7 downto 0));
44	end component;
45	
46	
47	signal a : std_logic_vector(3 downto 0);
48	signal b : std_logic_vector(3 downto 0);
49	signal c : std_logic_vector(7 downto 0);
50	

51	begin
52	
53	
54	lab3_wallance_test : lab3_wallace port map(
55	a => a,
56	b => b,
57	c => c
58	);
59	
60	
61	process
62	
63	begin
64	a <= "0101";
65	b <= "1010";
66	
67	wait for 10ns;
68	
69	a <= "0011";
70	b <= "1100";
71	
72	wait for 10ns;
73	
74	a <= "1010";
75	b <= "1100";
76	
77	wait for 10ns;
78	
79	a <= "0001";
80	b <= "1000";
81	
82	wait for 10ns;
83	
84	a <= "1110";
85	b <= "1011";
86	
87	wait for 10ns;
88	
89	a <= "1111";
90	b <= "1111";
91	
92	wait for 10ns;
93	
94	a <= "0000";
95	b <= "0000";
96	
97	wait for 10ns;
98	
99	a <= "1010";
100	b <= "1111";

101	
102	wait for 10ns;
103	
104	a <= "0001";
105	b <= "1111";
106	
107	wait for 10ns;
108	
109	a <= "1010";
110	b <= "1111";
111	
112	
113	wait;
114	
115	end process;
116	
117	
118	
119	end testing;

SCREENSHOT OF CODE

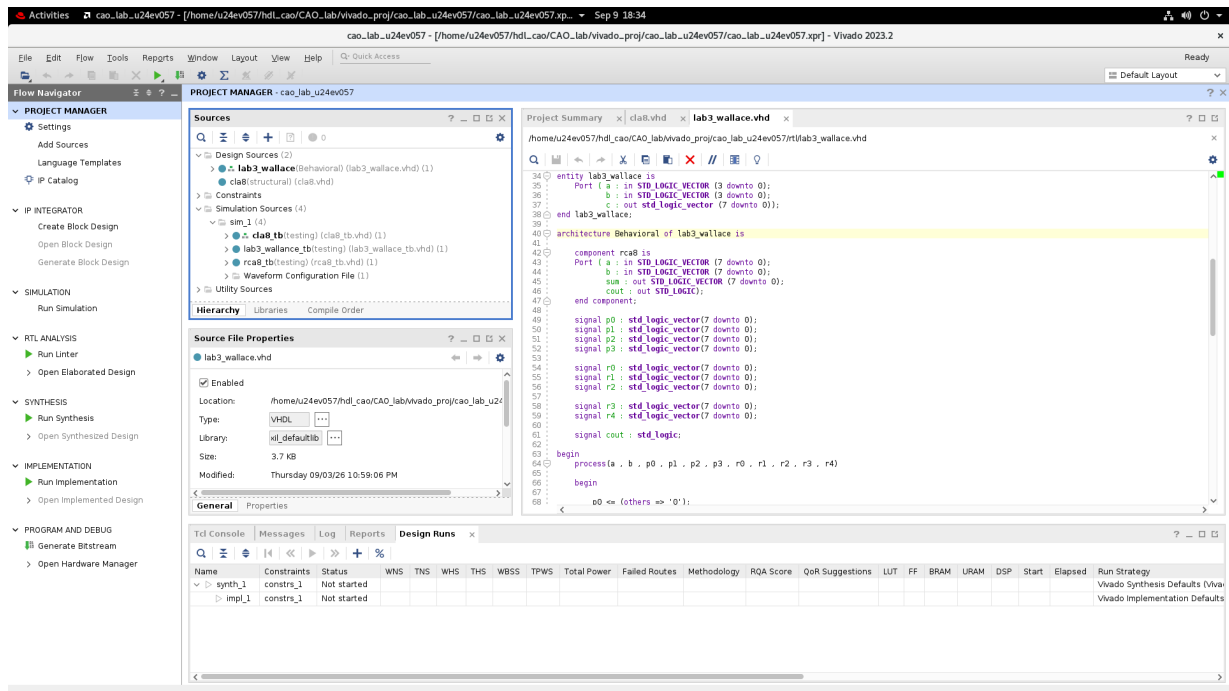


Figure 1: Design

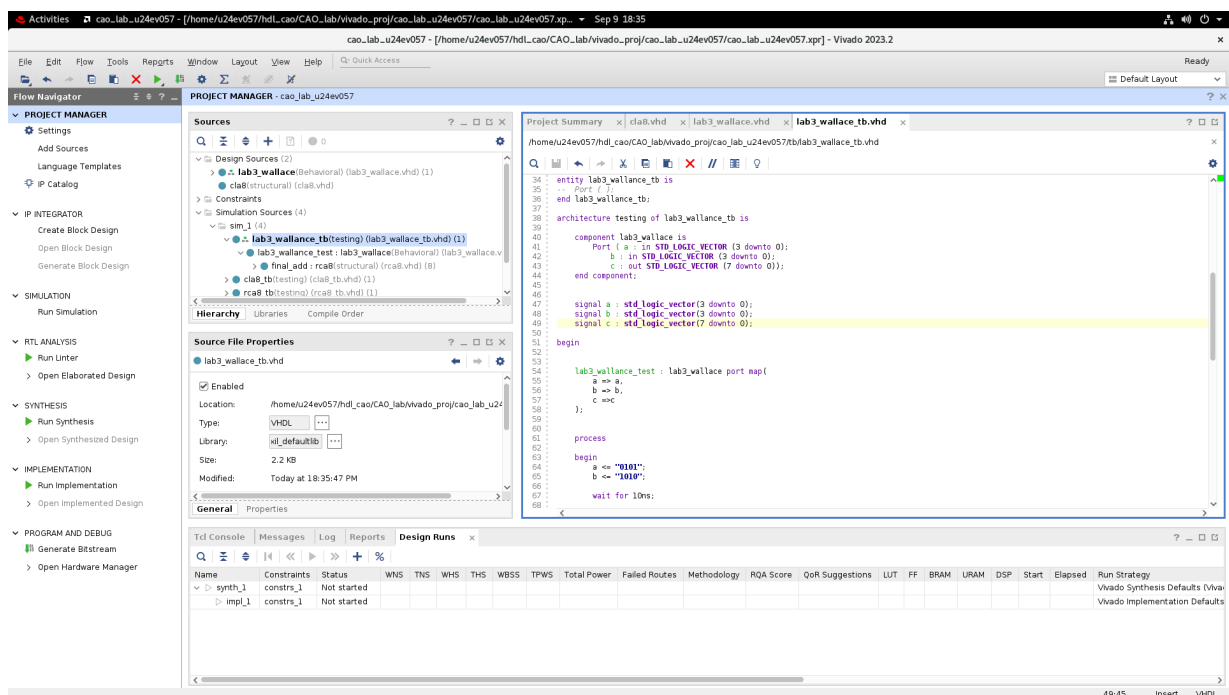


Figure 2: Tb

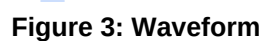


Figure 3: Waveform

